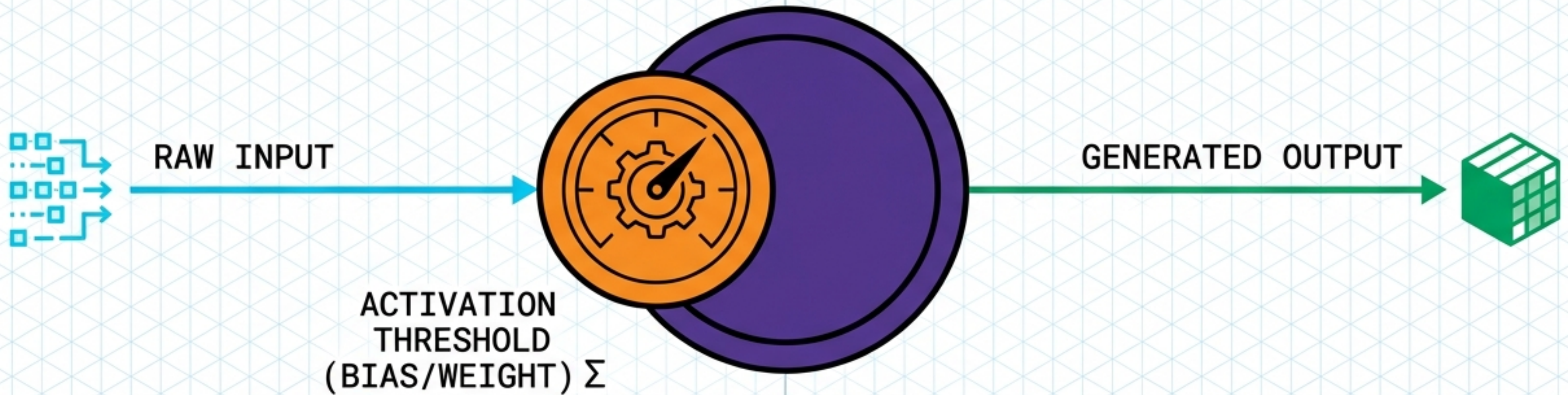


The Blueprint of Deep Learning

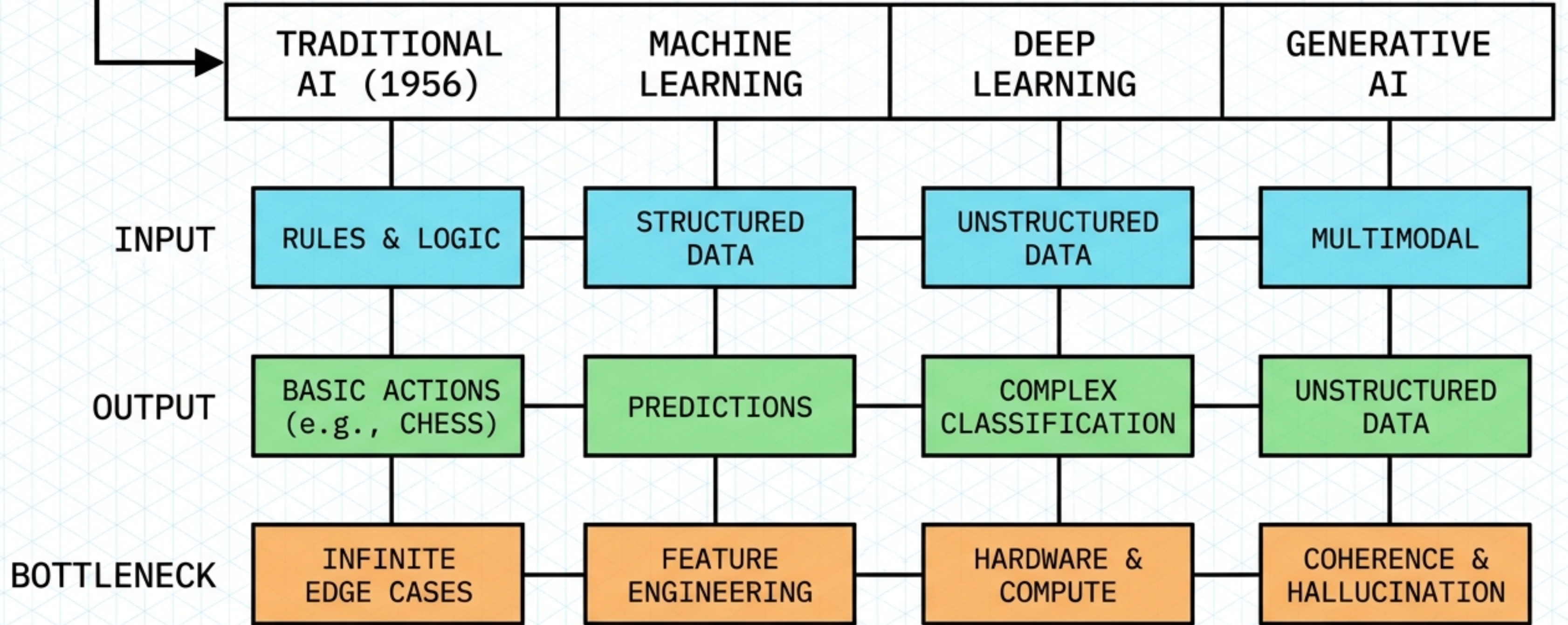


FILE: MIT-INTRO-NN // ALGORITHMIC ARCHITECTURE

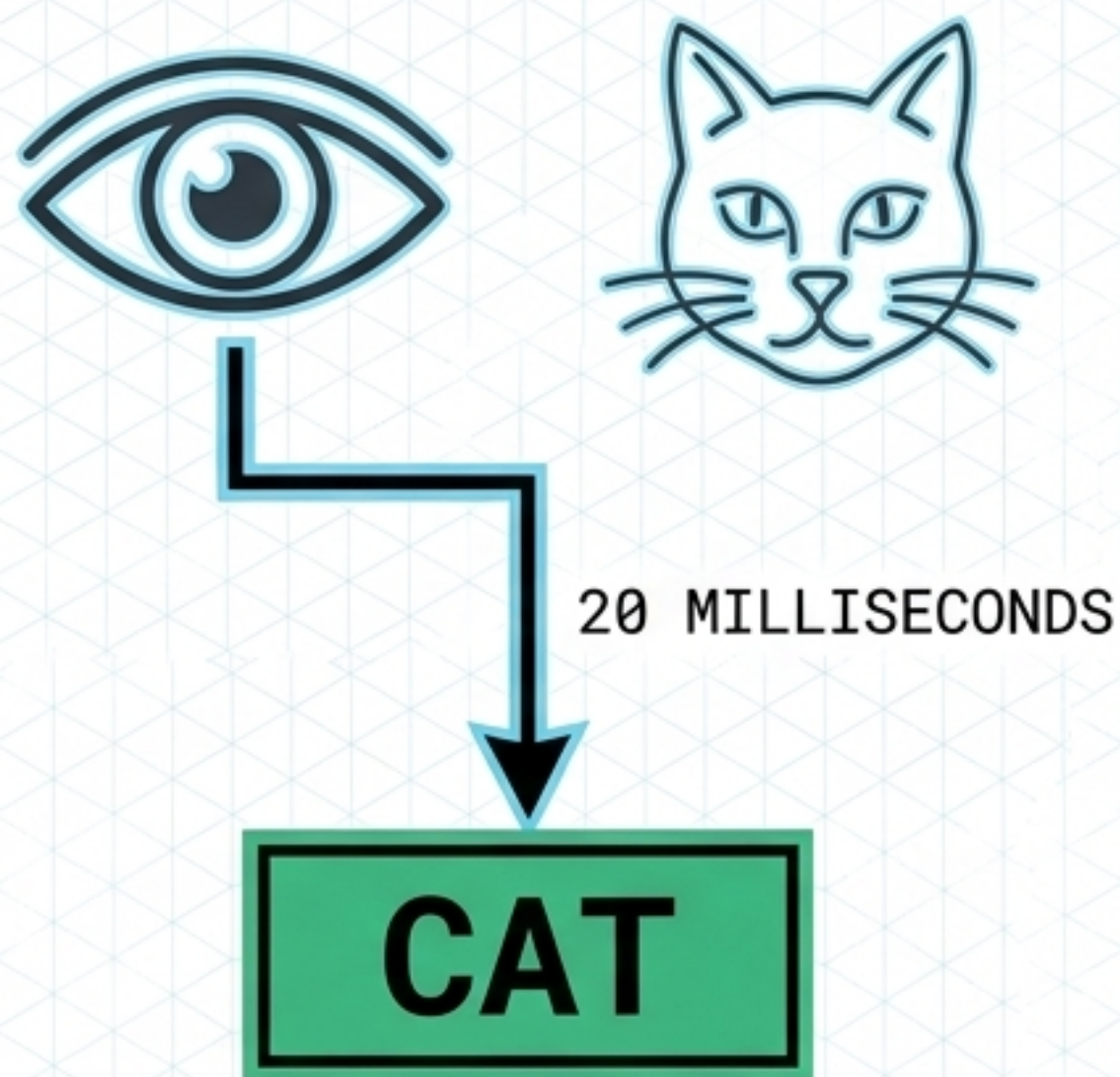
Mapping the mechanics from raw unstructured data to generative AI.

1956: Dartmouth conference founders (including MIT's Minsky, McCarthy, Shannon) expected to solve AI "by that fall."

THE FOUR ERAS OF ARTIFICIAL INTELLIGENCE

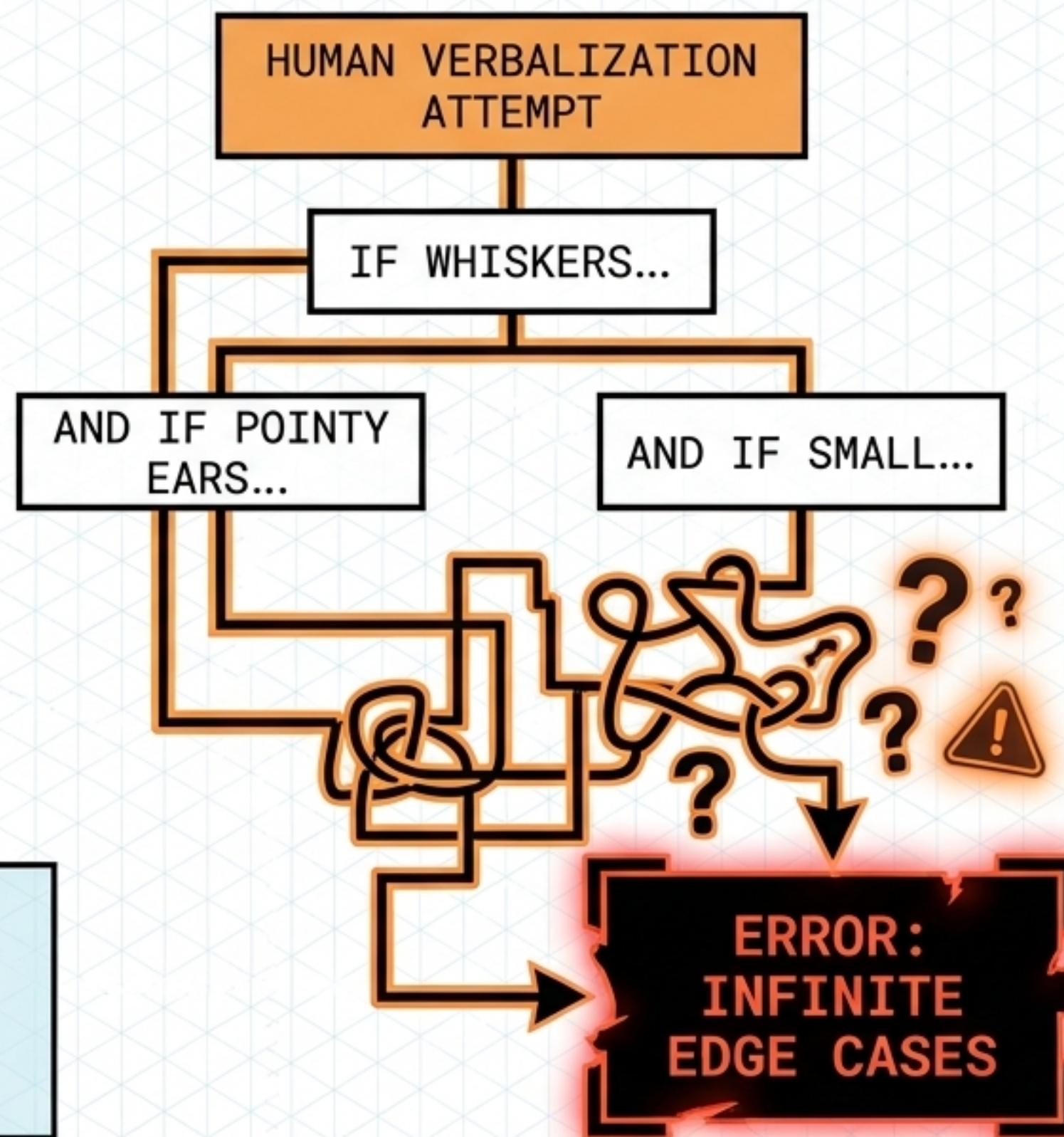


POLANYI'S PARADOX BROKE TRADITIONAL AI



WE KNOW MORE THAN WE CAN TELL.

Human articulation is incomplete. If we cannot explain how our brains recognize a cat in 20ms, we cannot write the if/then rules required for a computer to do it.



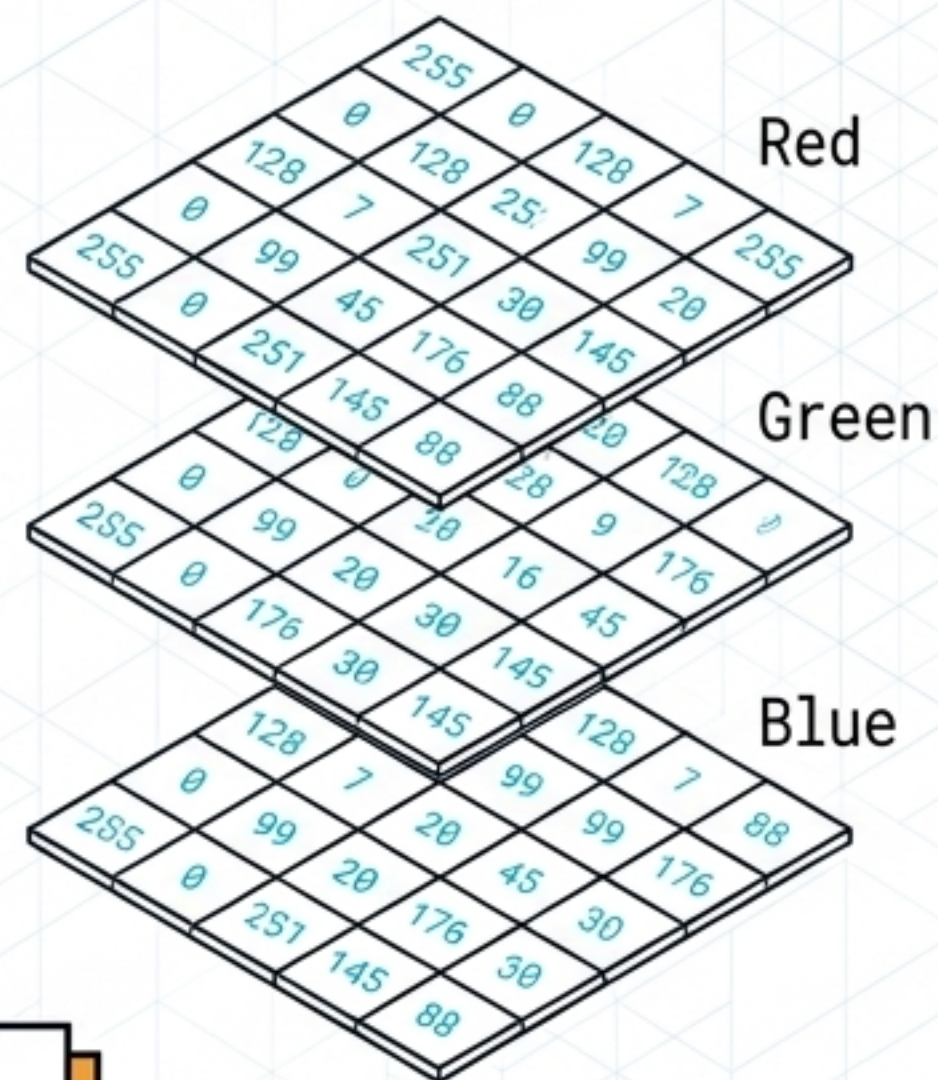
THE UNSTRUCTURED DATA DIVIDE

STRUCTURED DATA

	Age	Smoking Status
	45	Yes
	62	No

Clean, numerical, ready for traditional ML.

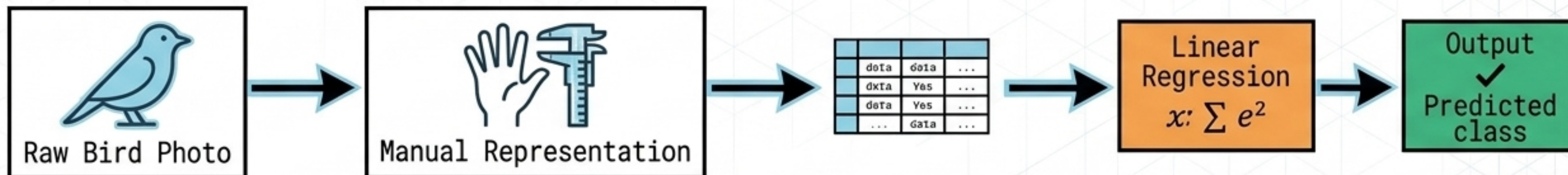
UNSTRUCTURED DATA



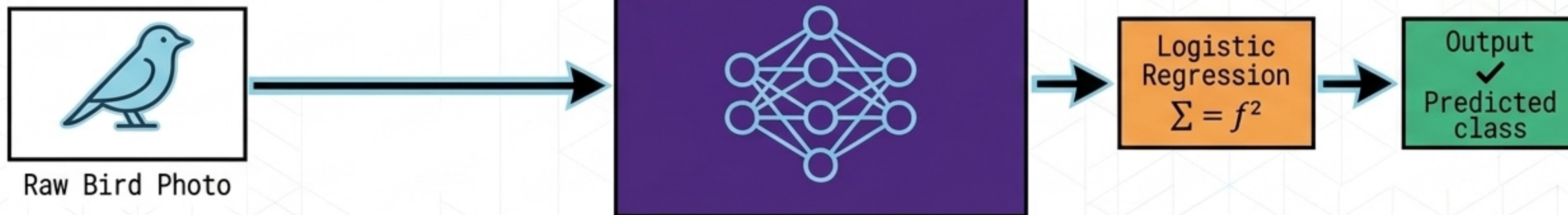
INSIGHT: Raw unstructured data has no intrinsic semantic meaning. The number 251 in the blue layer could mean a piece of sky, water, or blue paint.

REMOVING THE HUMAN BOTTLENECK

TRADITIONAL MACHINE LEARNING



DEEP LEARNING



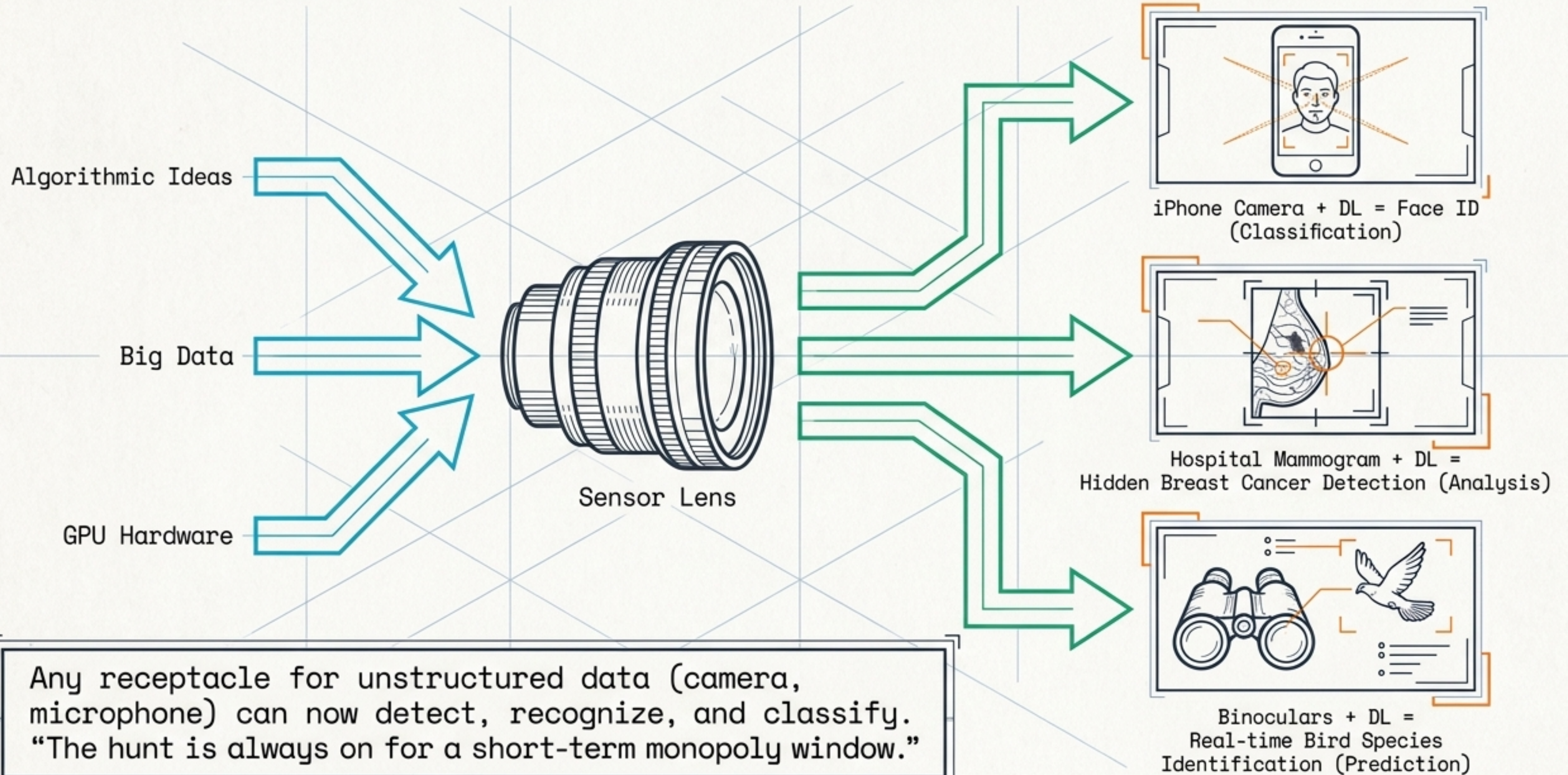
DEFINITION: REPRESENTATION

Translating raw data into a structured form algorithms can process.

THE BREAKTHROUGH

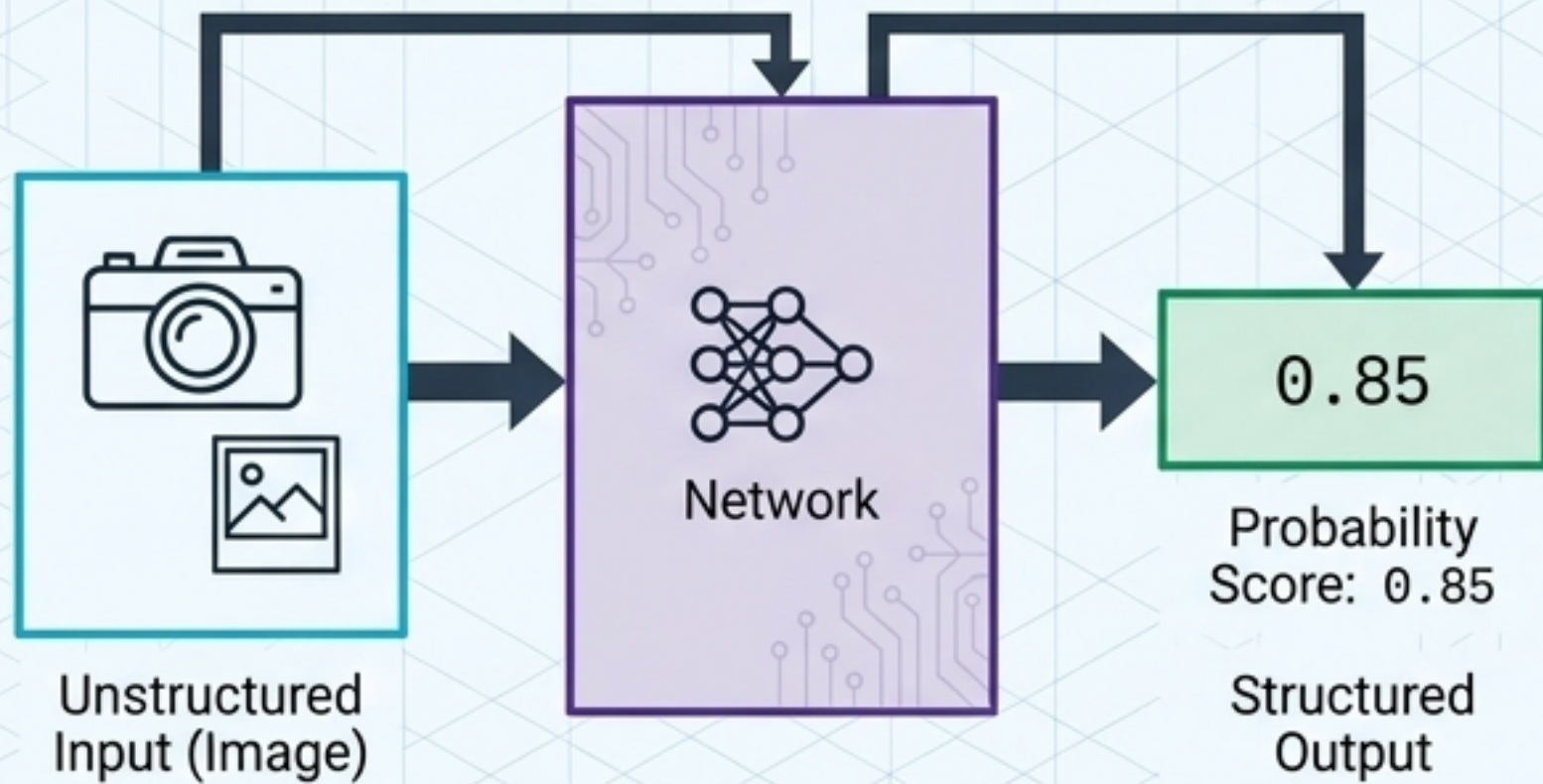
Deep Learning automatically extracts the right representations from raw input without human upfront processing.

TURNING EVERY SENSOR INTO A PREDICTIVE ENGINE



THE SHIFT FROM CONSUMING TO GENERATING

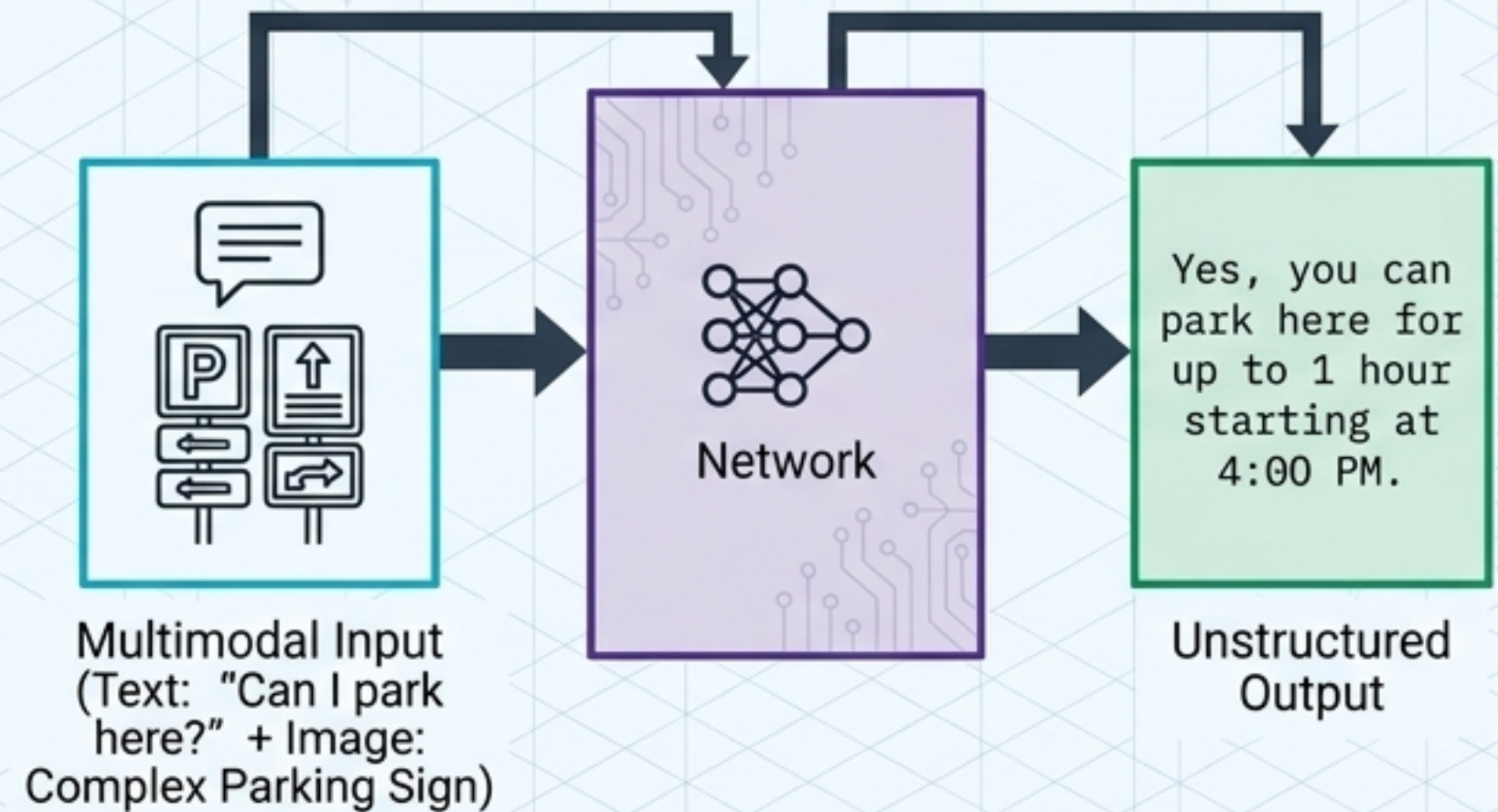
STANDARD DEEP LEARNING



SHIFT:

Historically, AI predicted a few numbers. GenAI produces entirely new unstructured data (images, text, music).

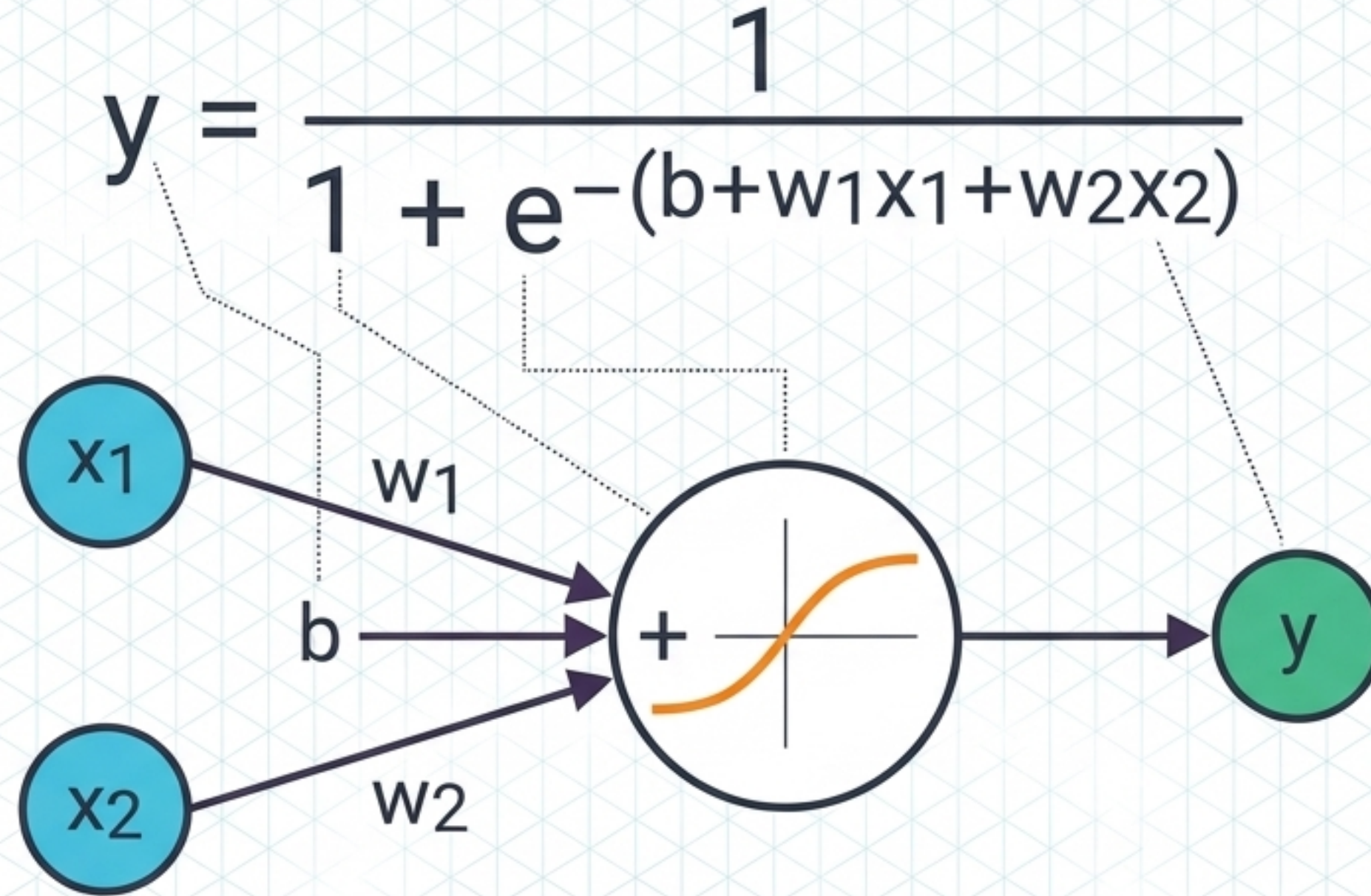
GENERATIVE AI



THE NEW NORM:

Multimodality (arbitrary sequences of text and image inputs) is replacing text-only models.

THE REGRESSION FOUNDATION



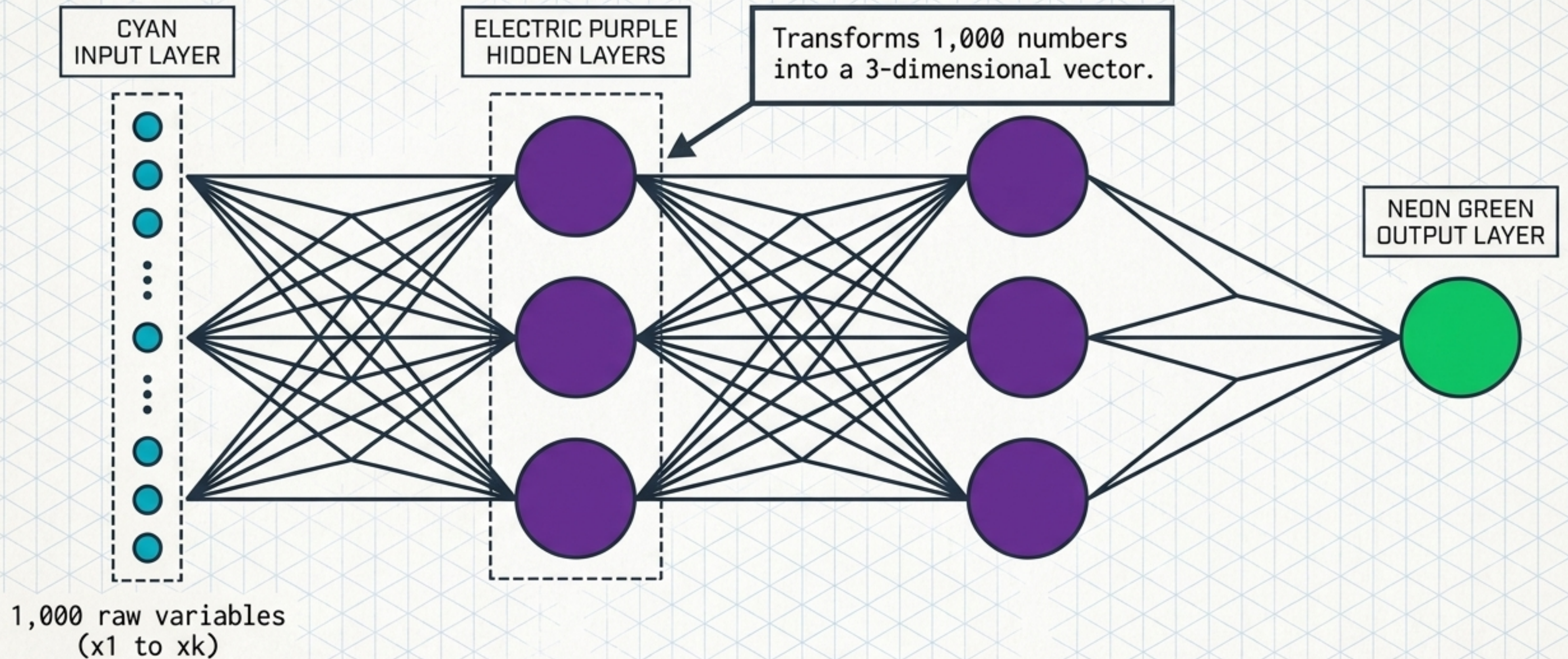
Logistic regression is essentially a neural network with zero hidden layers. It takes a vector of numbers and outputs a probability between 0 and 1.

THE TERMINOLOGY ROSETTA STONE

TRADITIONAL STATISTICS	DEEP LEARNING WORLD
Coefficients	Weights
Intercepts	Biases
Variables	Inputs
Logistic Regression	Zero-Hidden-Layer Network

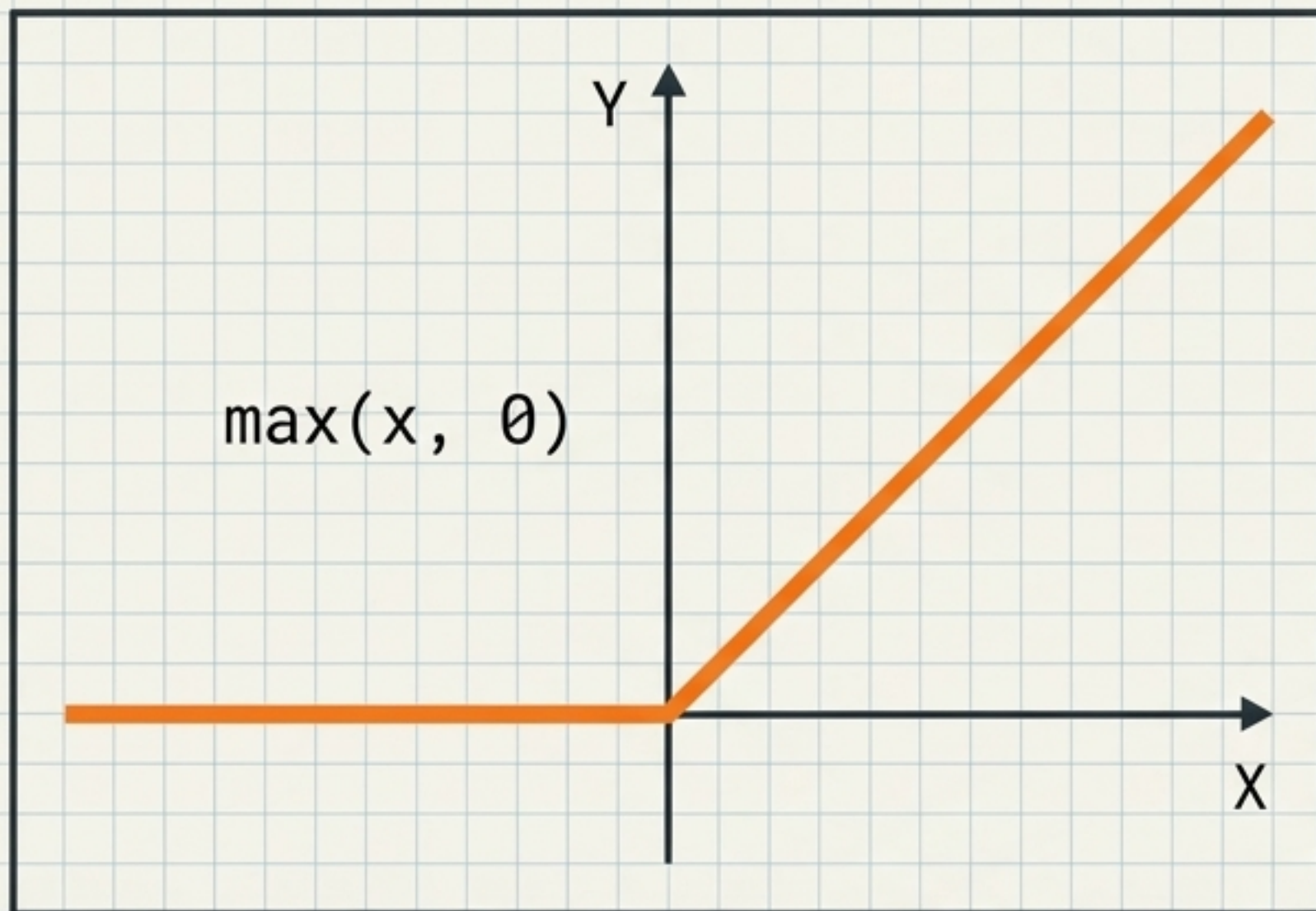
Context: When headlines claim a model's 'weights have been leaked,' it means the millions of learned coefficients and architecture have been exposed, allowing anyone to reconstruct the model.

STACKING LINEAR FUNCTIONS TO LEARN REPRESENTATIONS

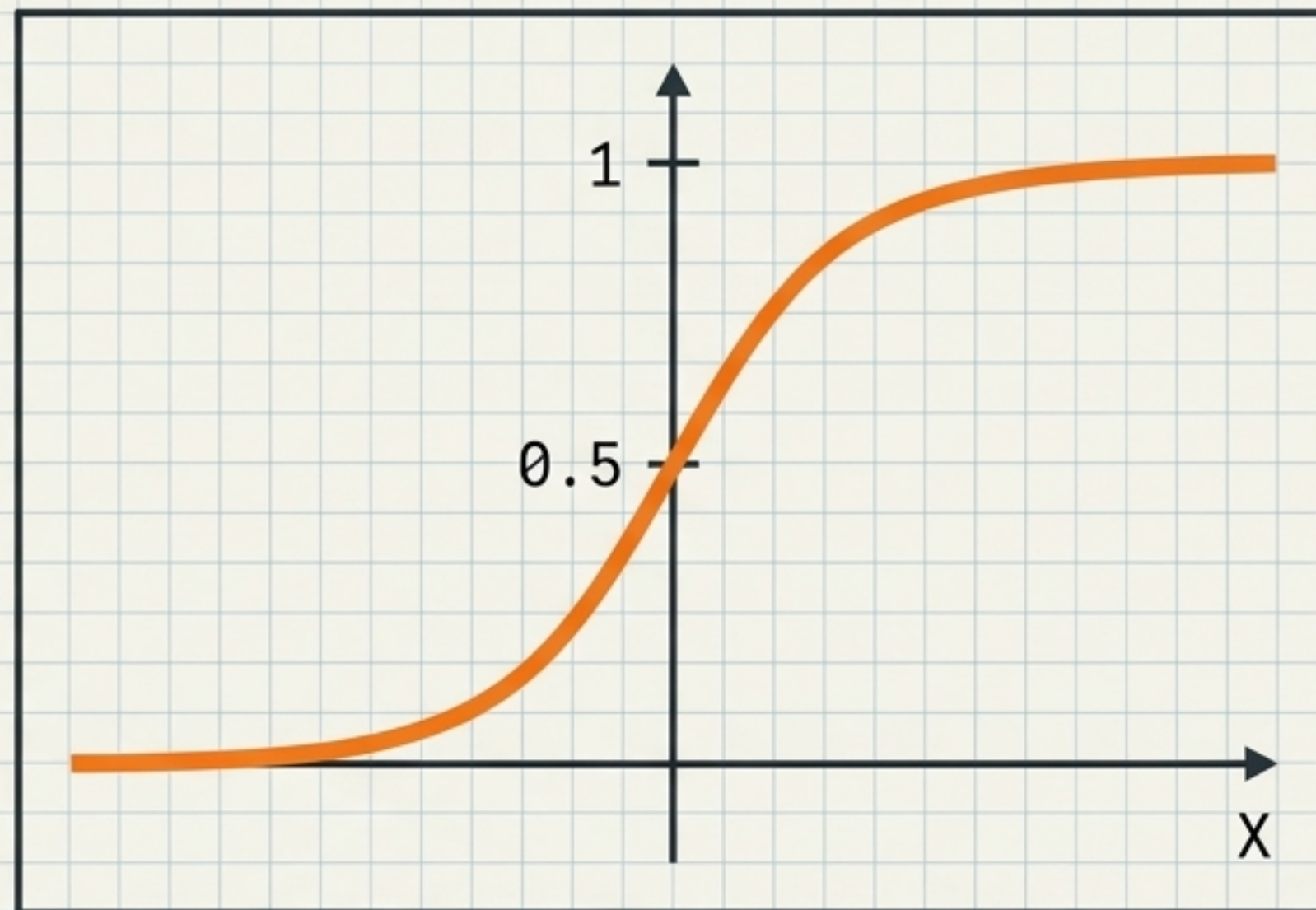


The Core Mechanic: Every hidden layer is a chance to mathematically transform the data. Deep networks repeatedly compress and transform inputs through stacked layers before feeding them to a simple final regression model.

THE NONLINEAR ENGINE: ACTIVATION FUNCTIONS



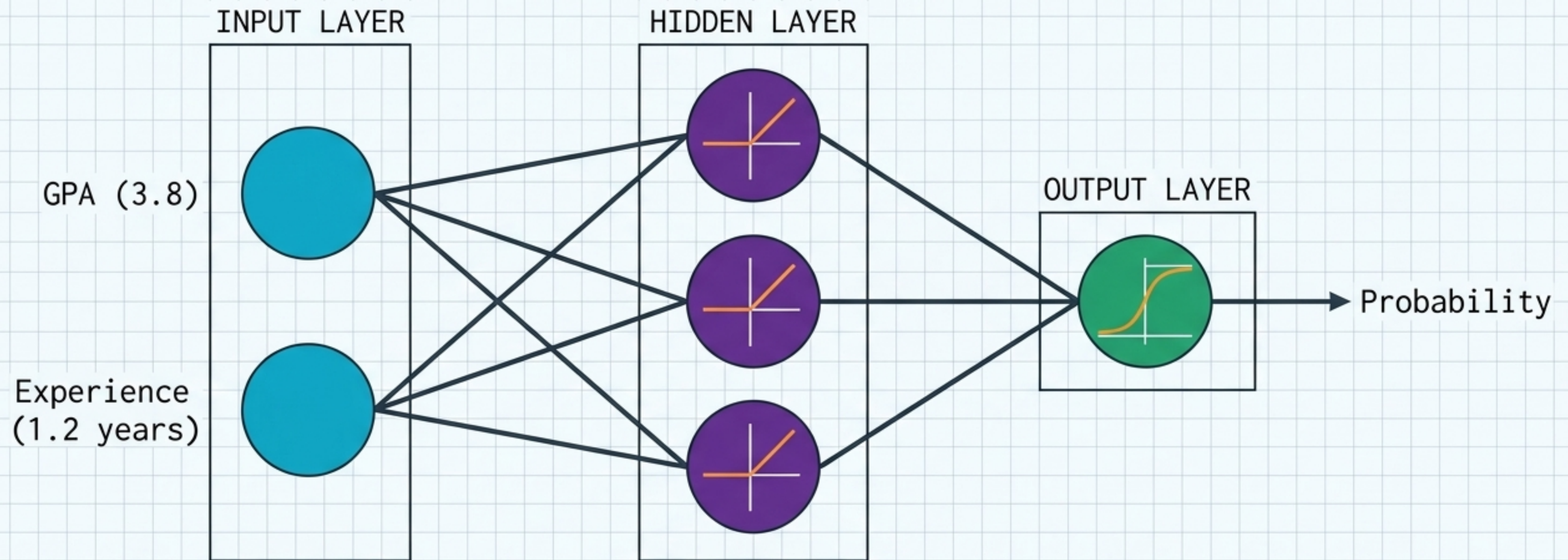
ReLU: The Hero of Deep Learning



Sigmoid: The Probability Output

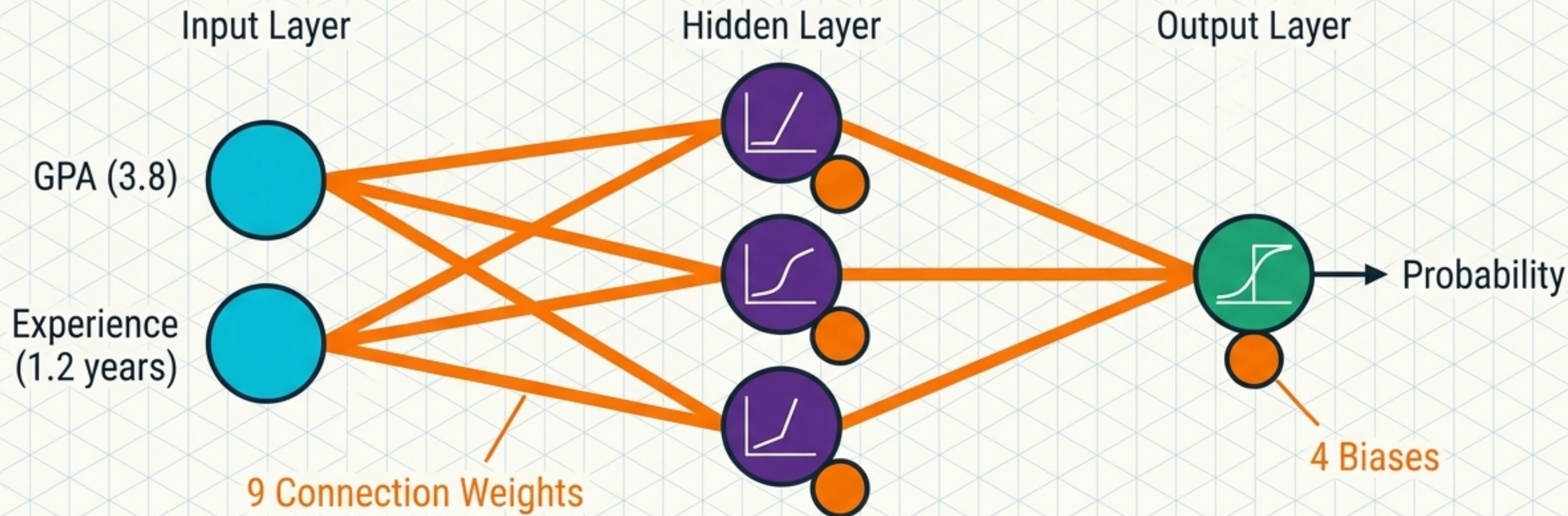
The Rule: Hidden layers require nonlinear functions (like ReLU) to learn complex patterns. If inputs are negative, ReLU zeroes them out (squashing noise). Output layers require functions that match the desired output (like Sigmoid for 0-to-1 probabilities).

BLUEPRINTING A COMPLETE NETWORK



- We dictate the architecture (inputs, outputs, hidden layers, nodes, and activation functions).
- The network mathematically figures out the internal weights during training to minimize prediction error.

AUDITING THE PARAMETERS



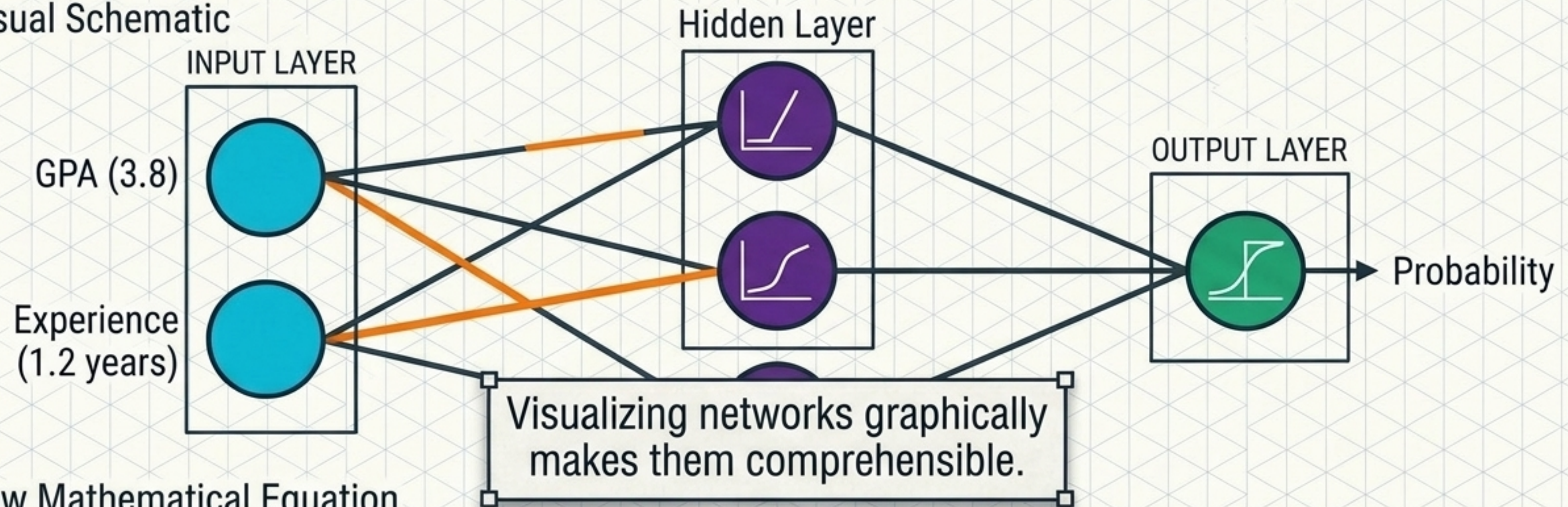
9 Weights + 4 Biases = 13 Total Parameters

Rule of Thumb: Every connection is a weight. Every node (after the input) requires a bias.

Modern models contain billions of parameters, but the fundamental math is identical to these 13.

The Source of the Magic: Complexity

Visual Schematic

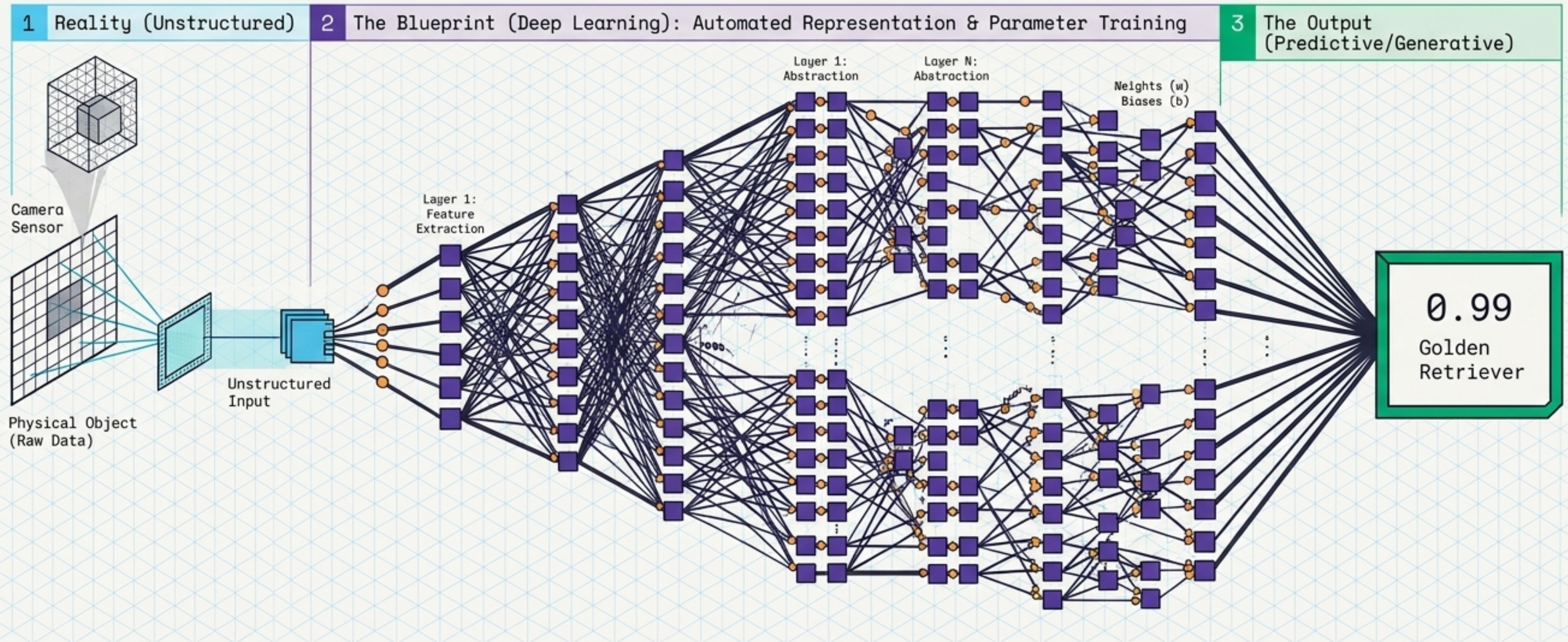


Raw Mathematical Equation

$$\text{Probability} = \frac{1}{1 + \exp\left(-\left(\frac{w_{o1} * \max(0, w_{h1_1} * x_1 + w_{h1_2} * x_2 + b_{h1}) + w_{o2}}{+ w_{o2} * \max(0, w_{h2_1} * x_1 + \dots * x_2 + b_{h2}) + w_{o3} * \max(0, w_{h3_1} * x_1 + w_{h3_2} * x_2 + b_{h3})} + \max(0, + w_{h3_2} * x_2 + b_{h3}) + b_o\right)\right)}$$

The “magic” of deep learning springs directly from this mathematical complexity. We can map complicated paths from just two inputs to a single output without manually programming the logic

The Architecture of Intelligence



Everything in modern AI stems from this architecture. By stringing together millions of simple linear transformations and nonlinear gates, we map raw reality to structured understanding.